

Quiz 5: Answer Key

Wednesday, March 4, 2026 5:05 PM

1 :	The BNF grammar rule $A \rightarrow \{ B \mid C \}$ means (Topic: EBNF Notation)
	A can be replaced by one B, followed by one C A can be replaced by zero or more BC pairs A can be replaced by zero or more Bs, followed by zero or more Cs You chose $\rightarrow$ A can be replaced by zero or more items, each of which may be B or C A can be replaced by one or more Bs, followed by one or more Cs
2 :	In a syntax diagram, a rectangle-shaped node indicates (Topic: Syntax diagrams)
	An error state An accepting state You chose $\rightarrow$ A label of another graph in the syntax diagram (like a nonterminal in a grammar) A literal symbol in the string being parsed (like a terminal in a grammar)
3 :	Two of the strings in the language of this grammar $S \rightarrow A A$ $A \rightarrow a c$ $A \rightarrow c a$ are (Topic: BNF Grammars)
	You chose $\rightarrow$ acca, acac aacc, ccaa a, c ac, ca ac, acacac
4 :	The BNF grammar rule $A \rightarrow B \{ C \}$ means (Topic: EBNF Notation)
	A can be replaced by one B, followed by one C A can be replaced by one or more Bs, followed by one or more Cs A can be replaced by either one B or one C You chose $\rightarrow$ A can be replaced by one B, followed by zero or more Cs A can be replaced by one B, optionally followed by one C
5 :	The BNF grammar rule $A \rightarrow \{ B C \}$ means (Topic: EBNF Notation)
	A can be replaced by either one B or one C <u>A can be replaced by one B, followed by one C</u>

	You chose → A can be replaced by zero or more BC pairs A can be replaced by one or more Bs, followed by one or more Cs A can be replaced by zero or more Bs, followed by zero or more Cs
6 :	BNF grammars and syntax diagrams are used to describe (Topic: BNF Grammars)
	the internal operation of the run-time stack during program execution the machine architecture used to implement programming languages You chose → the syntax of programming languages the semantics of programming languages
7 :	Two of the strings in the language of this grammar $S \rightarrow A S A$ $A \rightarrow c c$ $S \rightarrow d$ are (Topic: BNF Grammars)
	ccccd, ccd You chose → ccccdcccc, d ccdccd, ccd ccdddcc, cdc cc, dd
8 :	Two of the strings in the language of this grammar $S \rightarrow A A$ $A \rightarrow a a$ $A \rightarrow c c$ are (Topic: BNF Grammars)
	You chose → aacc, cccc acac, acacac a, c aa, cc acca, acac
9 :	A grammar is <i>ambiguous</i> if (Topic: BNF Grammars)
	there are two or more rules with the same right-hand side You chose → there is some string that can be derived from the grammar with two or more parse trees there are two strings generated by the grammar with the same parse tree there are two or more grammars that can generate the same language there is a rule missing from the grammar
10 :	The BNF grammar rule $A \rightarrow B C$ means (Topic: BNF Grammars)
	A can be replaced by one B, followed by zero or more Cs

	A can be replaced by one or more Bs, followed by one or more Cs You chose → A can be replaced by one B, followed by one C A can be replaced by either one B or one C A can be replaced by one B, optionally followed by one C
11 :	Two of the strings in the language of this grammar $S \rightarrow A B$ $A \rightarrow a a$ $B \rightarrow b \mid c$ are (Topic: BNF Grammars)
	aa, bc b, c You chose → aab, aac ab, ac aabc, abc
12 :	The start symbol of a grammar is one of the (Topic: BNF Grammars)
	Terminals Productions Parse trees You chose → Nonterminals
13 :	Two of the strings in the language of this grammar $S \rightarrow A B A$ $A \rightarrow c c$ $B \rightarrow d B$ $B \rightarrow d$ are (Topic: BNF Grammars)
	cc, dd ccdd, ddcc You chose → cccddcc, ccdcc cccc, cdc ccd, dcc
14 :	The BNF grammar rule $A \rightarrow B \mid C$ means (Topic: EBNF Notation)
	A can be replaced by one or more Bs, followed by one or more Cs A can be replaced by one B, followed by one C A can be replaced by one B, optionally followed by one C You chose → A can be replaced by either one B or one C A can be replaced by one B, followed by zero or more Cs
15 :	In a syntax diagram, a circle-shaped node indicates (Topic: Syntax diagrams)

You chose
→

A literal symbol in the string being parsed (like a terminal in a grammar)

An error state

A label of another graph in the syntax diagram (like a nonterminal in a grammar)

An accepting state

From <https://csweb.rowan.edu/~kilroy/provine/WebQuiz/review?quiz_try=30370>